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Feasibility study for simple on-line Raman spectroscopic detection of microplastic particles in water using perfluorocarbon as a particle-capturing medium

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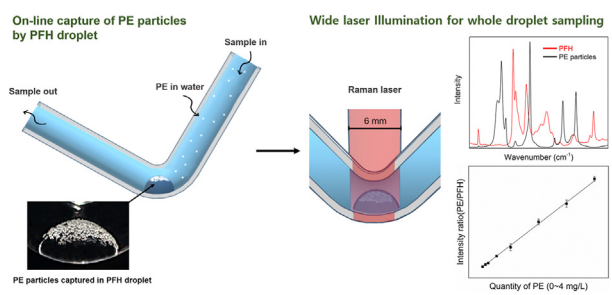
HIGHLIGHTS

- Raman spectroscopic on-line detection of microplastics (MPs) in water was demonstrated.
- This scheme uniquely utilized perfluorohexane (PFH, C_6F_{14}) as an MP-capturing medium.
- PFH droplet in a sampling tube effectively captured PE particles in tap water (recovery: 95.2%).
- All the captured PE particles were recognized by a wide area illumination Raman scheme.
- The measurement was quantitative in the analysis of PE particles in both tap and sea water.

GRAPHICAL ABSTRACT



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ABSTRACT

A simple Raman spectroscopic scheme for on-line detection of microplastics (MPs) in water is demonstrated. Instead of using a conventional physical filter for MP separation, perfluorohexane (PFH, C_6F_{14}) was deployed as an MP-capturing medium in this study. When PFH was added into a water-filled L-shape tube, it formed a firm droplet at the bottom of the 90° curve due to its strong hydrophobicity and high density. When a tap water sample containing dispersed polyethylene (PE) particles was flowed through the L-tube, the contained PFH droplet effectively captured the PE particles, with an average recovery of 95.9%. Next, for reliable quantitative analysis, it was necessary to measure the entire PE particle captured PFH droplet in Raman spectral acquisition without partial spectroscopic sampling. Therefore, a wide area illumination (WAI) scheme providing a laser illumination diameter of 6 mm was adopted for sampling of the whole droplet. The intensity ratios of PE and PFH peaks in the collected spectra clearly increased with elevated quantities of dispersed PE particles. When samples of PE particles were measured in sea water, which possesses much higher ionic strength than does tap water, the shapes of PE particle-captured PFH droplets did not change, and the accuracy was maintained. Based on these results, the demonstrated analytical scheme is feasible for field analysis; further study is required to strengthen its utility.

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1. Introduction

Microplastics (MPs) have been consistently found in oceans and

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rivers at considerable levels [1–5] and have become a threat to the surrounding ecosystems as well as to human health [6,7]. MPs in water have been analyzed using diverse analytical methodology. Micro infrared (IR) and Raman spectroscopy with prior filtration of MPs are among the most common methods [8–10] due to their capacity for identification of MPs as well as quantification. Filters of various materials have been used to separate MPs from water, including cellulose [11,12], membrane [11], silicon [13], polycarbonate [1,14], glass microfiber [15], and stainless steel [16]. Vibrational spectra of separated MPs on the filter were acquired using a microscope. An automatic mapping technique able to detect MPs in a given area has been employed for faster spectral acquisition to reduce total analysis time [17].

As the methodology progresses, a fast analytical method able to measure MPs in an on-line manner is demanding for real-time and on-site monitoring. To realize on-line analysis, filtration as well as time-consuming spectral acquisition steps should be simpler. Therefore, an alternative strategy enabling instantaneous capture of MPs dispersed in water along with their simultaneous one-step spectral acquisition is necessary. With this motivation, the current work explored potential Raman spectroscopy-based analytical schemes. First, instead of using conventional filters, perfluorohexane (PFH, C_6F_{14}) was used as an MP-capturing medium. PFH, in which the hydrogens in *n*-hexane are fully substituted with fluorine, is highly hydrophobic and is expected to form a droplet and effectively attract MPs dispersed in water. Since the density of PFH (1.67 g/mL) is much higher than that of water, the PFH droplet will settle and retain its shape in water without rupture. Similarly, hydrophobic mediums such as olive oil [18] and canola oil [19] were adopted for manual off-line separation of MPs in water, soil, and compost samples and subsequent recoveries ranged from 89.1 to 95.0%. Nonetheless, these edible oils are difficult to be employed for on-line capture of MPs in flowing water sample, since their droplet will move with the flowing water instead of stationing at a certain location for MP-capture owing to their lower density than that of water.

Second, a fast one-step spectral acquisition covering a whole MP-captured PFH droplet is essential for correct sample representation, ensuring reliable quantitative analysis. If only part of the MP-captured PFH droplet is sampled in spectral acquisition, accuracy of MP quantity determination inevitably degrades. To solve this problem, the size of MP-captured PFH droplet was maintained to slightly less than 6 mm, and a wide area illumination (WAI) scheme providing a laser illumination diameter of 6 mm was adopted to fully cover the MP-captured PFH droplet in Raman spectral acquisition.

To incorporate both PFH-based particle capture and WAI scheme, an L-shape tube was adopted to hold the PFH droplet in water at the bottom of the 90° curve as shown in Fig. 1. Using this arrangement, quantitative determination of polyethylene (PE) particles in water was attempted as a model study. Seven different masses of PE particles (0.1, 0.2, 0.5, 1.0, 2.0, 3.0, and 4.0 mg) were dispersed in tap water, and each PE particle-dispersed water sample was allowed to flow into the PFH droplet-containing L-tube. Raman spectra of the PE particle-captured PFH droplet were directly acquired using the WAI scheme. The variation of PE peak intensity according to PE particle quantity was examined, and subsequent performance of quantitative analysis and recovery of PE particle-capture were examined. Further, the scheme was evaluated for determination of PE particle quantity in sea water, a matrix with much higher ionic strength than tap water.

2. Experimental section

PE powder (density: 0.92 g/cm^3, purity: greater than 99.0%) and PFH were purchased from Sigma Aldrich. The average particle size of PE powder determined using a particle size analyzer (Malvern Instruments, Malvern, U.K.) was 303 ± 88.9 μm. Fig. 2 shows the microscopic image of randomly chosen PE particles. The sizes of some particles are indicated directly on the image. As seen, some of the particles are roughly spherical; while, the shapes of remained ones are diversely different without a common pattern. The number of particles counted in a 1 mg PE powder was 129.3 ± 10.3 based on 6 separate particle-counts of each new sample.

The L-shaped quartz tube (refer to Fig. 1) was prepared at a local glasswork shop and had an inner diameter of 6 mm. Local tap water (Seoul, Korea) was directly used, and sea water sampled at the eastern shore of the Korean peninsula was shaken 10 times before use. Initially, 150 μL PFH was introduced into the water-filled L-tube and formed a droplet at the curved bottom (Fig. 1). Although the PFH droplet could be stably retained in water due to its strong hydrophobicity and high density, a careful handling of PFH is necessary since it can cause skin, eye, and respiratory irritations as described by United States Environmental Protection Agency (US-EPA) [20] and corresponding Material Safety Data Sheet (MSDS) [21]. Next, a 1.0 L water sample containing dispersed PE particles passed through the PFH droplet-containing L-tube at a flow rate of 33 mL/min. The PE particle-dispersed water was continuously stirred during on-line measurement (30 min).

Next, from a PE particle-captured PFH (hereafter designated as PE-in-PFH) droplet, 130 μL PFH was drawn using a micropipette to reduce the total droplet volume to 20 μL and reduce the diameter of the PE-in-PFH droplet to slightly less than 6 mm. A wide area illumination (WAI) scheme providing a laser (785 nm) illumination diameter of 6.0 mm (PhAT probe, RamanRxn1 unit, Kaiser Optical Systems, USA) was used to fully cover the PE-in-PFH droplet at the curved-bottom for Raman spectral acquisition. Each spectrum (resolution: 4 cm^{-1}) was collected with laser exposure time of 5 s with 40 repeating scans. Asymmetric least-squares smoothing (AsLS) [22] was applied to correct baselines of acquired spectra using MATLAB version 7.0 (MathWorks Inc., MA, USA).

3. Results and discussion

3.1. Raman spectral features of PFH, PE particles, and PE particles captured in PFH

Fig. 3 (a) shows Raman spectra of pure PFH and PE particles in the 1600-900 cm^{-1} range. The spectral features of the samples considerably differ due to their dissimilar molecular structure. The PE peaks in the 1500-1400 cm^{-1} range, including 1458 (CH_2 rocking), 1440 (CH_2 bending), and 1416 cm^{-1} peaks, are separate from the peaks of PFH. Also, the 1128 cm^{-1} (C–C stretching) band presents with minimal influence from PFH peaks. PFH has a few strong peaks at 1380, 1209, and 972 cm^{-1}, which are minimally or not interfered by PE peaks. When Raman spectra of PE particles captured in PFH (PE-in-PFH) droplet are directly collected, the spectral features of PE and PFH appear together in the same trace. The 1440 cm^{-1} PE peak was chosen for quantitative analysis since it was second strongest among PE peaks, and there were no PFH peaks nearby to cause crosstalk. The most intense PE peak at 1295 cm^{-1} was excluded due to its overlap with nearby PFH peaks.

To examine the spectral features of the PE-in-PFH droplet, seven different quantities of PE particles (0.1, 0.2, 0.5, 1.0, 2.0, 3.0, and 4.0 mg) were added separately into 20 μL PFH droplets. Each PE-in-PFH was introduced into a water-filled L-tube made of quartz for Raman measurement. As demonstrated in previous publications, a glass container produced a broad Raman background over the 1800-1000 cm^{-1} range, and the background largely obscured sample peaks [23,24]. Therefore, a quartz tube that produces no

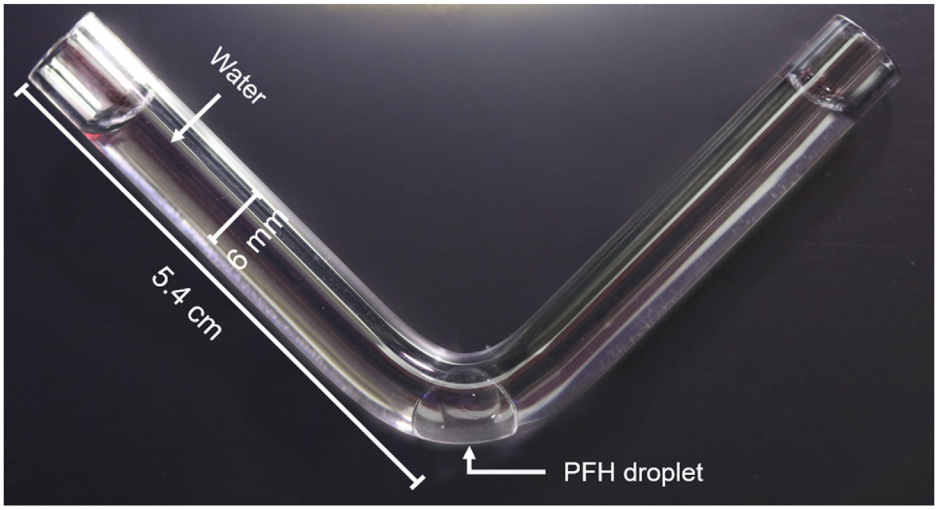

Fig. 1. Picture of water-filled L-tube containing a PFH droplet at the curved bottom.

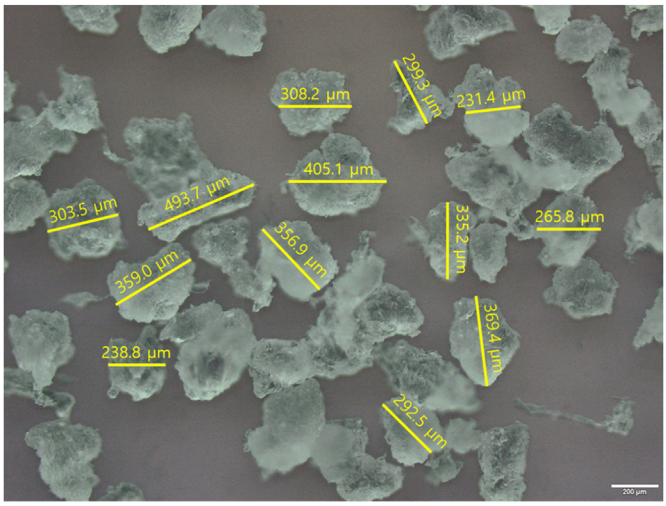

Fig. 2. Microscopic image of randomly chosen PE particles. The sizes of some particles are directly indicated on the image.

Raman background was used to secure accurate background-free analysis. The top left picture at the first row in Fig. 4 shows a PFH droplet positioned at the curved bottom of L-tube. The transparent droplet is clearly formed due to its high density and strong hydrophobicity. The width of the droplet was approximately 5.9 mm. The three pictures at the first row and four pictures at the second row show the PFH droplets in the water-filled L-tube that captured different amounts of PE particles. In all seven cases, the PE particles are clearly retained inside the PFH droplet by mutual hydrophobic interaction, even though the density of PE particles is lower than that of water. In the 0.1 mg PE-in-PFH droplet, the PE particles gather narrowly on the top of the PFH droplet. With increasing PE quantity (0.2 and 0.5 mg), the PE particles spread farther over the top surface of the PFH droplet. In the 1.0 mg PE-in-PFH droplet, the PE particles broadly populate the upper part of the PFH droplet. The PE particles distribute further down the internal surface of the droplet in the 2.0 and 3.0 mg PE-in-PFH droplets. In the case of 4.0 mg PE-in-PFH droplet, the particles cover most of the inner droplet surface. Since the sizes of PE-in-PFH droplets vary from 5 to 6 mm, the WAI scheme was suitable to cover the whole of each PE-in-PFH droplet in Raman spectral acquisition. When the amount of captured PE particles is large such as greater than 2.0 mg, the particles start to pack slightly inside the PFH droplet. Since the laser photons are able to diffuse (propagate) into the PE packing, the inner particles underneath the particles at the surface can be sufficiently sampled in Raman measurement. It is worthwhile to note that the packing thickness is not thick, as seen in the corresponding pictures.

Fig. 3 (b) shows Raman spectra of the 0.1, 0.2, 0.5, 1.0, 2.0, 3.0, and 4.0 mg PE-in-PFH droplets in the water-filled tube that are directly collected using the WAI scheme. Also, the spectrum measured from the blank PFH droplet without spiked PE particles is shown together. At a glance, the baselines of spectra are slanted due to the round droplet shapes, and the spectra also vary vertically since the shapes of PE-in-PFH droplets differ. For correction of baseline, AsLS smoothing was applied to each spectrum. Fig. 3 (c) shows the baseline-corrected spectra of the same samples. In the spectrum of blank PFH droplet, no noticeable PE peak at 1440 cm^{-1} is observed; while, the PFH peaks are apparent. In the spectrum of 0.1 mg PE-in-PFH droplet, the 1440 cm^{-1} PE peak starts to visually appear although its intensity is small. The PE peak becomes more intense with increasing the PE quantity to 0.2 mg. Afterward, the intensity of the PE peak at 1440 cm^{-1} clearly increases with increased quantity of PE particles in the PFH droplet. The baseline-corrected spectra were used for following analysis.

The use of absolute peak intensity may not be a desirable choice for quantitative analysis since laser power varies. Also, the slightly different shapes of PE-in-PFH droplets degrade the representation of the PE peak intensity to PE particle quantities. To overcome these problems, intensity ratio of PE to PFH peaks (designated as I_{PE}/I_{PFH}) was used for PE quantity determination, and the intensities of PE peak at 1440 cm^{-1} and PFH peak at 1380 cm^{-1} (designated by the vertical dotted lines in Fig. 3 (c)) were chosen for this purpose. Fig. 5 shows the variation of I_{PE}/I_{PFH} according to change of PE quantity in the PFH droplets (black squares). The calculated error bars were based on three independent PE-in-PFH droplet preparations and subsequent Raman measurements. The relationship is clearly linear (R^2: 0.998), as confirmed by the regression line. The use of I_{PE}/I_{PFH} enabled accurate measurement of the quantity of PE particles in the PFH droplet. The limit of detection (LOD) calculated based on 3 × standard deviation of the blank measurement was 0.09 mg. Since the average number of particles in a 1 mg of the sample was 129.3 ± 10.3 as described in the experimental section, the 0.09 mg

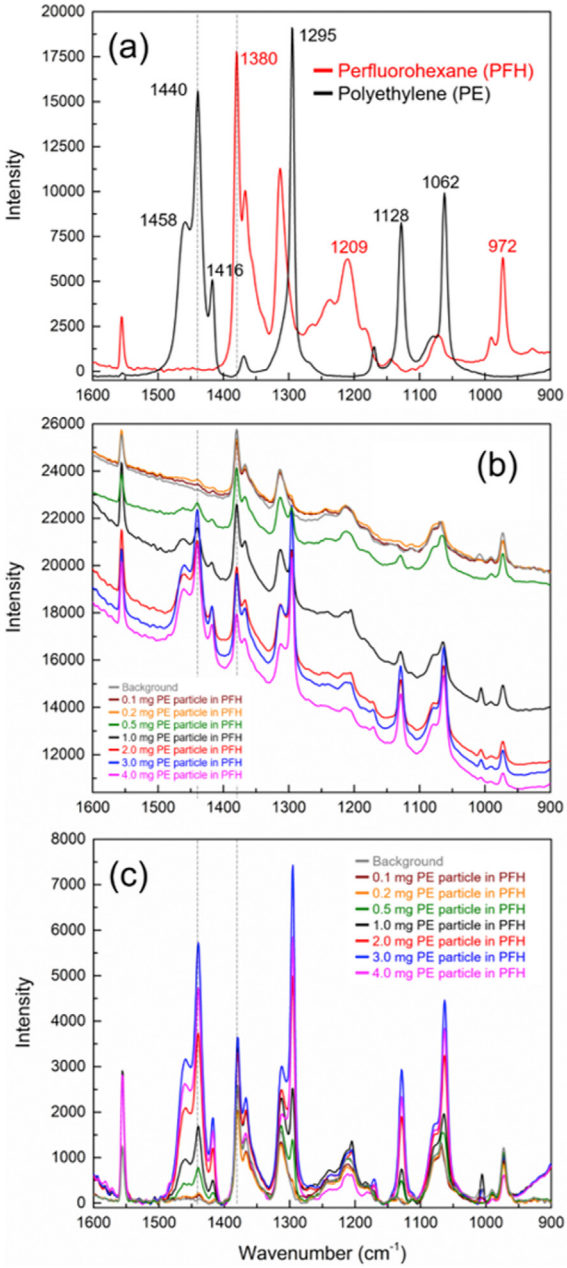

Fig. 3. Raman spectra of pure PFH and PE particles (a). Raman spectra of PFH droplets captured different amounts of PE particles in tap water (b) and corresponding baseline-corrected spectra by AsLS (c).

of sample (the minimum detectable quantity) was equivalent to approximately 12 particles.

3.2. On-line measurements of PE particles in tap water and sea water

For on-line analysis, 150 μL PFH was added to the tap water-filled L-tube. The 150 μL PFH volume allowed the PFH droplet to occupy most of the internal volume at the L-curve, creating a small sample flow channel above the droplet (refer to the leftmost picture at the third row in Fig. 4). This was performed to maximize collision of dispersed PE particles with the PFH droplet for efficient PE particle capture. If the volume of added PFH droplet is small, a larger proportion of particles would pass through the channel without contacting the PFH droplet, decreasing the efficiency of particle capture.

Next, 1.0 L tap water samples containing 1.0, 2.0, 3.0, and 4.0 mg dispersed PE particles were separately passed through the PFH-contained L-tube at a flow rate of 33 mL/min. Also, only a 1.0 L tap water was measured with the same fashion. The pictures of corresponding PE-in-PFH droplets after the on-line sample flow are shown in the third row in Fig. 4. As seen, there are no visually identifiable PE particles in the PFH droplet after the measurement of the tap water only. In the cases of measuring the PE particle-contained tap water samples, the PE particles are well captured inside the PFH droplets, and the population of captured particles becomes larger with increased concentration of PE particles dispersed in water. To quantitatively evaluate the PE particle-capturing ability, the recoveries in each measurement were calculated as shown in Table 1. For measurement, the PE-in-PFH droplet was removed from the L-tube. Then, PFH was thoroughly evaporated, and the remaining PE particles were weighed. For each sample, this procedure was independently repeated three times. The recoveries are good and range from 94.4 to 96.8% (average: 95.9%). This result confirms that the PFH droplet effectively attracts the dispersed PE particles and strongly retains them even under a continuous sample flow.

Since the PE-in-PFH droplets in the on-line measurement were larger than those of the 6 mm laser illumination, their sizes needed to be reduced for whole-droplet Raman sampling. Therefore, the total volume was decreased to 20 μL as described in the experimental section. Pictures of size-reduced PE-in-PFH droplets for each case are shown in the fourth row of Fig. 4. Raman spectra of the droplets were collected, and subsequent I_{PE}/I_{PFH} values were investigated in relation to dispersed PE particle quantity (refer to the red circles in Fig. 5). I_{PE}/I_{PFH} values vary in a linear fashion with the increase of PE particle quantity, while locate below the regression line in the reference measurement. This negative bias is equivalent to the lower PE peak intensity from the incomplete recovery of PE particles dispersed in water (Table 1). Considering the imperfect recovery, the accuracy of PE particle quantity determination in on-line measurement is superior.

Since MPs are found extensively in the sea, it is worthwhile to evaluate analytical characteristics of the proposed methods such as droplet shape and particle-capture ability in sea water, providing a much higher ionic strength than tap water. To do so, only sea water (1.0 L) was flowed into the PFH-contained L-tube, and the resultant PFH droplet (20 μL) was examined (refer to the leftmost picture of the bottom row in Fig. 4). As seen, a very small amount of unidentified organic matter in sea water was captured inside the PFH droplet; the droplet shape was not dissimilar to that of tap water. When the Raman spectrum acquired from the PFH droplet was analyzed, the spectral features corresponding to the captured organic matter were absent. Next, 1.0 L sea water samples containing 1.5, 2.5, and 3.5 mg dispersed PE particles were measured in the same on-line fashion. The pictures at the bottom row in Fig. 4 show the resulting PE-in-PFH droplets (20 μL). Again, the amounts of PE particles held in the droplets increase with elevated dispersed PE particle quantity in sea water, and the droplet shapes are similar to those in the tap water measurements. Fig. 5 shows the I_{PE}/I_{PFH} values calculated from the obtained Raman spectra of three PE-in-PFH droplets in sea water (blue triangles). The I_{PE}/I_{PFH} values are closely located along the bottom regression line, indicating that the accuracy of particle quantity determination is not influenced by the surrounding medium, whether tap or sea water.

4. Conclusions

The capture of water-dispersed PE particles using PFH and

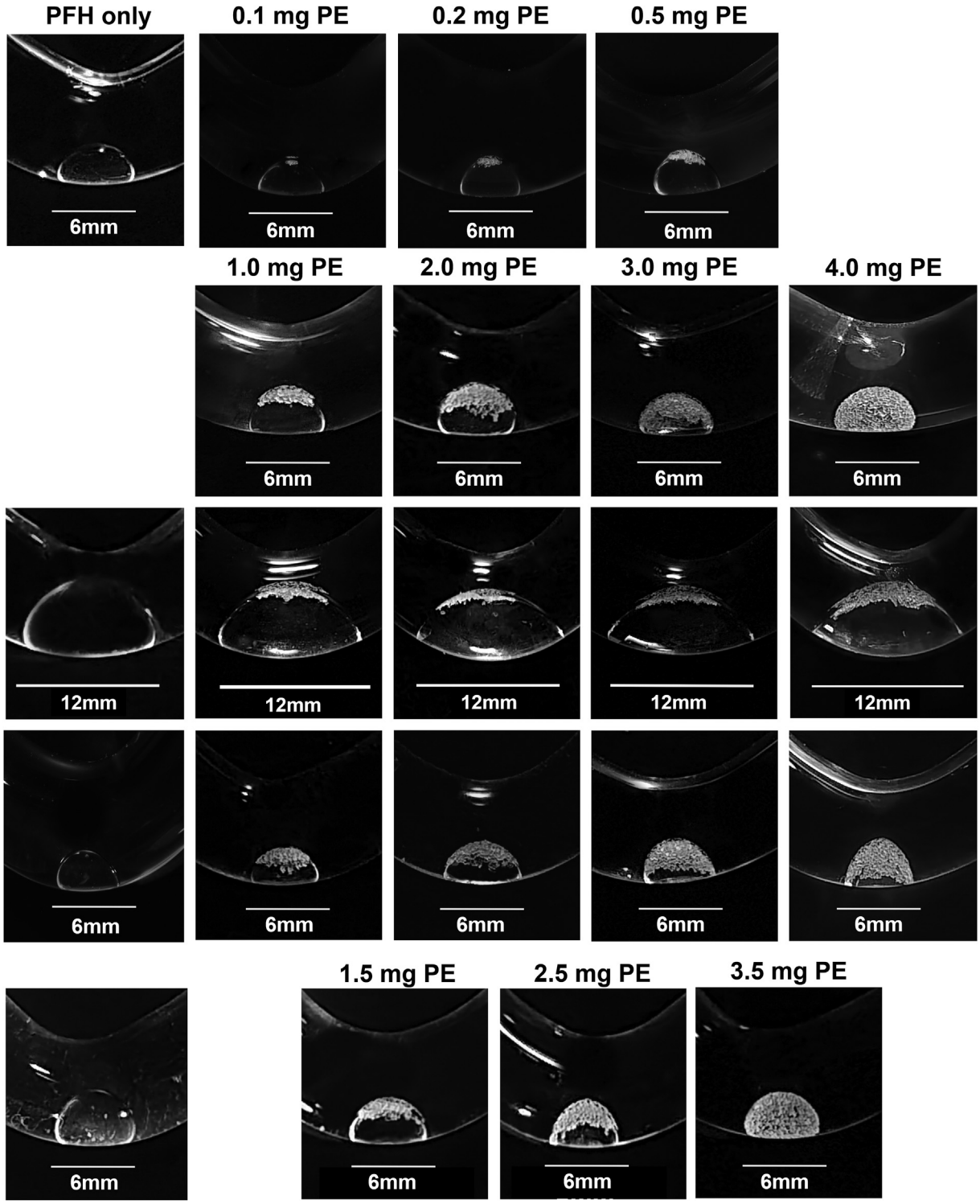

Fig. 4. Pictures of 20 μL PFH droplets that captured different amounts of PE particles in the reference measurement (first and second row), and 150 μL (third row) and 20 μL (fourth row) PFH droplets that captured PE particles in tap water samples. Pictures of 20 μL PFH droplets that captured PE particles in sea water samples are also shown (fifth row). PFH-only droplets from each measurement are shown at the left of each row.

subsequent Raman spectral acquisition of the whole PE-in-PFH droplet by the WAI scheme is viable for simple on-line determination of PE particle quantity in water. The water-immiscible, dense PFH droplet possessing strong hydrophobicity effectively captured the dispersed PE particles in water (average recovery: 95.9%). Although the shapes of PE-in-PFH droplets were not consistent, the Raman sampling of whole droplets by the WAI scheme was helpful to secure accurate quantitative analysis. As reported in former publications [2–5,10], the numbers of detected items (particles) in a cubic meter of the field samples varied from 10 to 4320; while, the

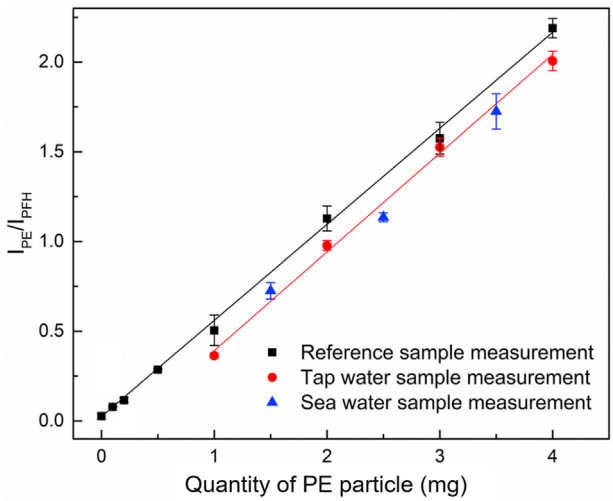

Fig. 5. Relationships between PE particle quantity and I_{PE}/I_{PFH} resulting from the reference measurement (black squares), measurement of tap water samples (red circles), and measurement of sea water samples (blue triangles). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1
Recoveries of PE particle quantities captured in three replicates of PFH.

Added PE particle quantity	Recovered PE particle quantity			Recovery
1.00 mg	0.93 mg	0.97 mg	0.97 mg	95.7 ± 1.8%
2.00 mg	1.81 mg	1.95 mg	1.91 mg	94.4 ± 3.0%
3.00 mg	2.89 mg	2.99 mg	2.81 mg	96.5 ± 2.5%
4.00 mg	3.98 mg	3.99 mg	3.65 mg	96.8 ± 4.0%

minimum detectable number of PE particles was around 12 particles in this study. Therefore, the proposed analytical method is potentially applicable for analysis of field sample when a cubic meter of sample is measured.

Beyond this feasibility study, several issues should be addressed for field application of the demonstrated analytical method. First, design of a new flow cell able to enhance particle-capture efficiency up to nearly 100% is necessary. Direct introduction of water sample into PFH droplet is among the possible solutions. Second, there are different types of MPs such as polypropylene (PP) and polyethylene terephthalate (PET) particles; therefore, particle-capturing characteristics of PFH for diverse MPs need to be studied, though it is reasonable to expect most MPs to be attracted to PFH droplets in water. Third, when different types of MPs are captured in a PFH droplet, identification and quantification of separate MPs will be demanded. For this purpose, sophisticated chemometric analyses such as multivariate curve resolution [25,26] are necessary. Fourth, a strategy for prior handling of micro-organisms and sediments present in field water is needed. These issues are under investigation in our research group.

CRediT authorship contribution statement

Sanghoon Cho: Methodology, Software, Validation, Formal analysis, Resources, Data curation, Writing – review & editing, Visualization. **Yunjung Kim:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – review & editing, Visualization. **Hoeil Chung:** Conceptualization, Methodology, Resources, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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